Serosurvey of recent measles infection in region D2, Denden, December 2026

Reported Sunday 10 January 2027. Data: *Serosurvey, region D2, 14-18 Dec 2026* (`serosurvey_denden_D2_2026-12.csv`, 9 rows). *Exercise material: fictional countries, simulated data.*

Denden's public health institute tested blood samples from a random sample of the residents of region D2, Vejager and its surroundings, for antibodies from a recent infection with the measles variant. The file has one row for each 10-year age group.

How the survey was done

- **Who:** 300 people in each 10-year age group, from 0-9 to 80 and over, 2,700 in all. They were drawn at random from D2's population register, each person independently, and everyone selected gave a sample.
- **When:** samples were taken from Monday 14 to Friday 18 December 2026.
- **The assay** tells antibodies from a recent infection with the variant apart from antibodies left by vaccination or by an earlier measles infection. **Assume it is perfect:** no false positives and no false negatives.
- **Timing:** the antibodies become detectable about two weeks after infection, around the time of the rash, and stay detectable. Someone infected in the two weeks before their sample tests negative.
- Infections count whatever their course: with or without symptoms, in vaccinated and unvaccinated people.

The columns

column	meaning
`age_group`	10-year age group, written "0 to 9", "10 to 19" and so on, and "80+"
`tested`	people tested in the age group
`seropositive`	people with antibodies from a recent infection
`population`	people living in D2 in the age group, from the population register

Reading the file

- **Only region D2 was sampled.** The other regions of Denden, and the other countries, are not covered.
- **The same number was tested in every age group,** not in proportion to the population. To estimate the share of all D2 residents infected, weight each age group by its population.
- **The results are not linked to the line list.** No one in the survey can be matched to a notified case.

Not in the file

- **Vaccination status, symptoms, sex and household** of the people tested.
- **The test date of each sample.** Only the week is known.